Konsider Phase 6C Prompt — Higher-Education Opportunity Criteria and Evidence Exploration

Role

Work as the lead product, education-data, research-evidence, and geography engineer for the Konsider repository.

This is a research and product-definition phase only.

Do not implement runtime schemas, production workers, ranking changes, active releases, API changes, UI changes, preference presets, or applicant-admission logic.

Temporary research scripts, retained local source captures, deterministic institution mappings, and committed research reports are allowed where necessary.

Dependency

Proceed after Phase 6B has frozen the common Opportunity Criterion meaning and tri-state contract.

Reuse these public states:

- `VERIFIED_STRONG_SIGNAL`
- `STRONG_SIGNAL_NOT_ESTABLISHED`
- `INSUFFICIENT_EVIDENCE`

Reuse the Phase 6B asymmetric evidence rule:

- a positive strong signal may be established through one approved high-confidence path or a frozen combination of paths;
- a negative result requires sufficiently complete assessment;
- an incomplete assessment must remain `INSUFFICIENT_EVIDENCE`.

If Phase 6B changes the names of the public states, use the owner-approved final names consistently.

Background

The prior Phase 6A study found:

- UIS field-of-study graduate shares but no compatible graduate counts;
- only 75–76 fresh countries for the engineering and ICT graduate-share indicators;
- ShareAlike obligations requiring explicit treatment;
- no pinned OpenAlex capture;

- no retained ROR capture;
- no deterministic institution-to-Konsider-locality mapping;
- and no measured country coverage or overlap analysis for the engineering academic ecosystem.

Those findings blocked weightable ordering criteria.

This phase asks a different question:

> Can Konsider verify that a country contains a strong higher-education ecosystem in a particular field, without claiming admission access, affordability, or applicant fit?

New product concept

A higher-education Opportunity Criterion is a non-ranking filter.

A user may eventually request:

> Only consider countries where Konsider has verified a strong higher-education opportunity in engineering.

The filter determines eligibility. Existing ordering criteria determine rank among eligible countries.

The opportunity result does not imply:

- admission probability;
- international-student eligibility;
- tuition affordability;
- scholarship availability;
- accreditation recognition;
- language suitability;
- visa access;
- teaching quality at every institution;
- quality of every programme;
- or personal fit.

Objective

Identify a small, user-comprehensible portfolio of higher-education Opportunity Criteria for which strong positive and negative evidence can be established for a practically useful portion of the stable 91-country universe.

The phase must determine:

1. Which education fields deserve public filters?
2. What does “strong higher-education opportunity” mean for each field?

3. Which source combinations can establish institution quality, breadth, research activity, or field specialisation?
4. Can at least 60 of 91 countries be assessed honestly?
5. Are countries likely to appear in user shortlists adequately covered?
6. Can absence from a ranking or source be interpreted safely?
7. Which candidates should proceed to a future implementation-design phase?

Do not implement the filters in this phase.

Starting candidate portfolio

Research the following candidate Opportunity Criteria.

Prefer a small number of broad, recognisable fields over numerous thin programme categories.

Core candidates

1. **Higher-education opportunities — Engineering and technology**

2. Intended meaning: the country contains a substantial and demonstrably strong engineering and technology higher-education ecosystem.

3. **Higher-education opportunities — Computer science and ICT**

4. Intended meaning: the country contains a substantial and demonstrably strong computing or ICT higher-education ecosystem.

5. **Higher-education opportunities — Medicine and health sciences**

6. Intended meaning: the country contains a substantial and demonstrably strong medical or health-sciences higher-education ecosystem.

7. Do not imply that foreign medical qualifications will be recognised.

8. **Higher-education opportunities — Business and finance**

9. Intended meaning: the country contains a substantial and demonstrably strong business, management, accounting, economics, or finance higher-education ecosystem.

10. Narrow the field if source taxonomies cannot support the combined public name honestly.

Additional candidates to probe

1. **Higher-education opportunities — Natural sciences**

2. Intended meaning: the country contains a substantial and demonstrably strong natural-sciences higher-education ecosystem.

3. **Higher-education opportunities — Broad university excellence**

4. Intended meaning: the country contains meaningful breadth of internationally prominent higher-education institutions across fields.

5. Evaluate whether this is useful or too broad and redundant with the field-specific filters.

Do not automatically split undergraduate and postgraduate opportunity into separate public criteria.

First determine whether exact sources can distinguish programme level reliably. If not, keep the public construct at general higher-education ecosystem level and expose level limitations in evidence.

Construct discipline

The initial construct should be:

> Does this country contain a sufficiently strong and sufficiently broad higher-education ecosystem in the selected field to constitute a verified opportunity signal?

This is different from:

- number of students;
- percentage of graduates in the field;
- research output alone;
- presence of one institution;
- general university reputation;
- quality of every programme;
- international-student access;
- or applicant admission probability.

The research must decide which evidence dimensions are necessary and sufficient.

Evidence dimensions to explore

A. Field-specific institution prominence

Research exact, legally usable, reproducible sources for field-specific institution rankings or classification bands.

Possible source families may include:

- publisher field/subject rankings;
- official ranking data downloads;

- transparent ranking tables;
- globally scoped field-specific university lists;
- discipline-specific accreditation or recognition lists where they measure the intended construct.

For every candidate source verify:

- complete list versus displayed subset;
- rank or band methodology;
- field taxonomy;
- institution count;
- country coverage;
- publication year;
- reuse and derivative rights;
- reproducible acquisition;
- whether absence has a meaningful interpretation.

A country absent from a top-100 list may be assessed only for:

presence of an institution in that top-100 list.

It must not automatically be assessed as lacking good education.

B. Institution breadth

Evaluate whether strong opportunity requires more than a single institution.

Possible measures:

- number of institutions within approved ranking bands;
- number of distinct strong institutions;
- breadth across locations;
- breadth across subfields;
- concentration in one institution versus a wider ecosystem.

Avoid arbitrary top-100 cliffs.

Test multiple bands, such as:

- top 100;
- top 200;
- top 300 or ranked band equivalents.

Do not treat rank 99 and rank 101 as categorically different without sensitivity analysis.

C. Field-specific research ecosystem

Research OpenAlex and other production-compatible scholarly evidence.

If using OpenAlex:

- choose and freeze an acquisition route;
- pin exact filters, taxonomy version, dates, cursor/pagination policy, and response hashes;
- freeze the recent multi-year window;
- use ROR or another approved canonical institution registry;
- retain fractional attribution where appropriate;
- retain institution-country identity;
- distinguish research scale, breadth, and impact;
- avoid raw citation totals as the sole measure;
- review service terms for the chosen acquisition route.

Research evidence may support higher-education ecosystem strength, particularly at postgraduate and research levels. It must not be described as undergraduate teaching quality.

D. National education specialisation and capacity

Revisit UIS and other official education sources.

Evaluate:

- field graduate shares;
- field graduate counts;
- total graduates;
- enrolment by field;
- tertiary population denominators;
- field specialisation;
- compatible reference years;
- freshness;
- classification versions;
- licensing;
- coverage.

Do not combine graduate shares with enrolment totals from a different population.

Use field shares only as specialisation evidence, not as capacity.

Search for exact alternative official sources that provide compatible counts and totals.

E. Programme and level evidence

Explore whether reproducible global or multi-country sources can establish:

- programme existence;
- undergraduate versus postgraduate level;
- institution and campus;
- field;

- current offering status;
- accreditation or recognised-awarding status.

Do not rely on manual web searches of university pages as a production-global evidence path unless a bounded, reproducible shortlist method is explicitly being tested.

A programme-level path may be suitable later for shortlisted countries even if it cannot support a 60-country Opportunity Criterion.

F. Geographic accessibility

Determine whether the public opportunity filter needs locality evidence immediately.

For institution-level evidence:

- retain exact institution identity and country;
- retain city or locality when reliably available;
- evaluate ROR location data;
- evaluate multi-campus institutions;
- retain unmapped or conflicting locations;
- avoid nearest-city guesses;
- distinguish country-level opportunity from locality-level accessibility.

The Phase 6C research may approve a country-level education opportunity without full integration into the Phase 5 selected-locality universe, provided the public construct is country presence and no locality claim is made.

Do not block a valid national institution-presence signal merely because it cannot yet identify the best city.

However, record whether locality mapping will be required for future product explanation.

Composite evidence routes

For each candidate, evaluate a small number of explainable evidence routes.

Examples to test—not assumptions to adopt—include:

Route 1: exceptional institution presence

One or more institutions cross a very high field-specific prominence threshold.

Route 2: ecosystem breadth

Several institutions cross a broader credible threshold, demonstrating that opportunity is not concentrated in one institution.

Route 3: research and institution combination

Moderate institution prominence plus substantial recent field-specific research scale and breadth.

Route 4: education specialisation support

Institution or research strength plus credible national evidence of substantial field specialisation or graduate capacity.

Do not build an opaque composite score.

A country may receive `VERIFIED_STRONG_SIGNAL` through one frozen strong route or a defined combination of moderate routes.

Negative-state rule

`STRONG_SIGNAL_NOT_ESTABLISHED` may be assigned only when the approved evidence process has sufficient recall to detect a strong ecosystem.

Examples of potentially adequate negative assessment:

- an exhaustive or sufficiently broad field-specific institution universe was assessed;
- the country's institutions were mapped successfully;
- the research capture covered the country;
- the approved strong-signal routes were evaluated;
- none crossed threshold.

Examples requiring `INSUFFICIENT_EVIDENCE`:

- the ranking publishes only a limited visible subset with unknown omissions;
- the country is missing from OpenAlex/ROR mapping;
- field taxonomy cannot be mapped;
- the source year is stale;
- the country has unresolved institution identity conflicts;
- the source licence prevents retained evidence;
- only UIS graduate share is available;
- or one major evidence path is absent and negative assessment would be unsafe.

Minimum research gate

A candidate may be recommended for implementation only if all gates pass.

1. Assessable-country floor

At least **60 of the stable 91 countries** must be classified as either:

- `VERIFIED_STRONG_SIGNAL`; or
- `STRONG_SIGNAL_NOT_ESTABLISHED`.

`INSUFFICIENT_EVIDENCE` does not count.

2. Decision-relevance coverage

Using only existing published ordering criteria, use the same benchmark top-20 country shortlists frozen in Phase 6B.

Report assessment coverage for each education candidate.

The preferred gate is at least **16 of 20 assessable countries** in each relevant shortlist, especially:

- general balanced;
- family and education-oriented;
- governance and safety-oriented;
- affordability-sensitive; and
- career-prioritised where family education remains material.

3. Geographic and economic breadth

Assessable countries must not be limited primarily to wealthy English-speaking destinations.

Report coverage by:

- region;
- income group;
- language group where relevant;
- and size of higher-education system.

4. Discrimination

Report counts for:

- verified strong signal;
- strong signal not established;
- insufficient evidence.

Test whether the filter meaningfully narrows choices.

As a normal guide, the smaller assessed class should include at least eight countries.

Flag:

- saturation;
- one-country dominance;
- extreme concentration in Europe/North America;
- and sensitivity to minor rank-band changes.

5. Construct integrity

The public name must match the evidence.

Examples:

- research strength is not teaching quality;
- ranking presence is not admission accessibility;
- graduate specialisation is not institution quality;
- institution count is not programme breadth;
- top-university presence is not affordability;
- country presence is not locality access.

6. Source and legal gate

For every source freeze:

- exact asset or deterministic query;
- publisher and distributor;
- release or extraction date;
- methodology;
- field taxonomy;
- institution universe;
- licence and service terms;
- normalized-derivative rights;
- attribution;
- retained-raw policy;
- checksum or capture identity;
- refresh cadence.

Explicitly address:

- UIS ShareAlike obligations;
- ranking-table reuse and derived-country aggregation rights;
- OpenAlex data versus API-service terms;
- ROR and GeoNames attribution;
- and any third-party methodology restrictions.

Ambiguous production reuse blocks implementation recommendation.

7. Reproducibility

The approved evidence process must reproduce:

- source capture identity;
- institution matching;
- country mapping;
- field mapping;
- evidence components;
- status assignment;
- reason codes;
- coverage;
- shortlist analysis;
- and threshold sensitivity.

Institution identity and country mapping

Use stable institution identifiers wherever possible.

For every institution retain:

- source institution ID;
- canonical institution ID;
- display name;
- institution type;
- country;
- location evidence;
- field evidence;
- rank/band or research evidence;
- multi-campus status;
- mapping confidence;
- and rejection reason where applicable.

Do not merge similarly named institutions without deterministic evidence.

Do not assign a multi-country institution entirely to headquarters when the source evidence is campus-specific.

Required analyses

For every candidate report:

- public construct;
- user decision value;
- exact source assets;
- legal conclusion;
- field-taxonomy mapping;

- institution coverage;
- country coverage;
- shortlist coverage;
- region and income-group coverage;
- evidence-route pass rates;
- strong-signal examples;
- assessed-no-strong-signal examples;
- insufficient-evidence examples;
- threshold sensitivity;
- top-100/top-200/top-300 sensitivity where applicable;
- one-institution versus ecosystem-breadth sensitivity;
- research-output sensitivity;
- country-size correlation;
- source overlap;
- false-positive risks;
- false-negative risks;
- maintenance burden;
- refresh plan;
- and applicant-context boundary.

Explicitly test countries that are important relocation or education destinations but were problematic in Phase 6A.

Career and education portfolio relationship

Compare the approved or proposed education Opportunity Criteria with Phase 6B career Opportunity Criteria.

Report:

- where a country has strong education but weak career evidence;
- where it has strong career evidence but weak education evidence;
- where both are strong;
- where one or both are insufficient;
- whether field names align across the two portfolios;
- and whether shared taxonomy IDs would be useful in a later implementation.

Do not implement shared runtime taxonomies yet.

The report may recommend future shared IDs, such as:

- `technology`
- `science_engineering`
- `healthcare`
- `business_finance`
- `natural_sciences`

But do not add production contracts in this phase.

Decision categories

Assign exactly one disposition per candidate:

`APPROVE_FOR_IMPLEMENTATION_DESIGN`

The construct and tri-state evidence policy clear the research gates.

`APPROVE_WITH_NAMING_OR_SCOPE_CHANGE`

A narrower field, evidence kind, degree level, or public name is supportable.

`HOLD_SOURCE_GAP`

The construct is valuable, but exact sources, coverage, licensing, institution identity, or reproducibility are insufficient.

`HOLD_POLICY_OR_THRESHOLD_GAP`

Sources exist, but the strong/no-strong decision rule or negative-state completeness is not defensible.

`RESEARCH_SHORTLIST_ONLY`

The evidence can enrich a user's shortlist or country detail page but cannot support a 60-country Opportunity Criterion.

`REJECT_AS_OPPORTUNITY_CRITERION`

The candidate is too broad, too thin, too redundant, too source-dependent, or too close to applicant admissions.

Required deliverables

Create:

1. `docs/research/phase6c-education-opportunity-study.md`
2. `data/reports/phase6c-<date>/education-candidate-matrix.json`
3. `data/reports/phase6c-<date>/education-country-opportunity-evidence.jsonl`
4. `data/reports/phase6c-<date>/education-source-matrix.json`
5. `data/reports/phase6c-<date>/institution-mapping.jsonl`
6. `data/reports/phase6c-<date>/education-shortlist-coverage.json`
7. `data/reports/phase6c-<date>/education-threshold-sensitivity.json`
8. `data/reports/phase6c-<date>/career-education-crosswalk.json`
9. `data/reports/phase6c-<date>/approved-opportunity-portfolio.json`

10. deterministic research scripts and fixtures needed for replay.

The final approved portfolio should consolidate all Phase 6B and Phase 6C candidates recommended for a future implementation-design phase.

Do not include held or rejected candidates in the implementation portfolio.

Explicit non-goals

Do not:

- change the active release;
- publish a new release;
- modify affinity scoring;
- modify the PCC policy;
- modify the stable 91-country universe;
- add production Opportunity Criterion schemas;
- add filters to the API;
- add UI controls;
- add conversational behavior;
- implement institution recommendations;
- implement admission matching;
- implement fees or affordability;
- implement visa eligibility;
- implement qualification recognition;
- implement professional licensing;
- or weaken existing ranking criteria.

Verification

Run all gates relevant to the changed research artifacts.

At minimum:

- deterministic source replay;
- capture checksum verification;
- institution and country mapping reconciliation;
- exactly 91 country outcomes per candidate;
- JSON/JSONL schema validation;
- shortlist reconciliation;
- threshold-sensitivity replay;
- no-diff regeneration of deterministic reports;
- documentation-link validation;
- linting and formatting for scripts;
- existing tests affected by research and documentation changes.

Do not claim remote CI unless it has run.

Stop condition

Stop after the education research and consolidated portfolio decision are complete.

Report:

- education candidates researched;
- approved public names;
- evidence routes;
- assessable-country counts;
- shortlist coverage;
- strong/no-strong/insufficient counts;
- source and legal blockers;
- institution-mapping limitations;
- career/education crosswalk;
- final implementation-design portfolio;
- held and rejected candidates;
- files changed;
- tests run;
- commit SHA;
- and owner decisions required before any implementation phase.

Do not proceed to code, data model, contracts, releases, API, UI, or conversational integration.